ROSLYN WILLOUGHBY

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EDUCATION

PSL Research University **Paris, France**
Graduate coursework, AI & Society master's program
Studied mathematics for AI, and AI ethics and governance.

Washington State University **Pullman, WA**
BA, Social Science *Dec 2025*
GPA: 3.75/4.00, magna cum laude
Relevant Coursework: Probability & Statistics, Python Programming, Statistics in Psychology

RESEARCH INTERESTS

Trust in AI among domain experts, accessible and assistive AI, and AI governance.

RESEARCH EXPERIENCE AND TRAINING

WSU / NIH Training in Gerontechnology **Pullman, WA**
Trainee *January 2026 - February 2026*
Program led by Dr. Diane Cook & Prof. Maureen Schmitter-Edgecombe

- Studied interdisciplinary methods from computer science and neuropsychology that combine smart-environment and wearable sensor data with clinical cognitive assessments to detect functional decline linked to cognitive impairment earlier and more reliably.
- Trained in Python (pandas, scikit-learn) activity recognition methods for detecting cognitive decline from smart-home sensor data and wearable technologies.

Cervical Cancer Risk Prediction on Imbalanced Clinical Data **Pullman, WA**
Student Researcher *June - August 2025*
Advisor: Azam Boskabadi | Python, scikit-learn

- Reviewed Naive Bayes applications in healthcare, including cancer survival modeling.
- Cleaned and analyzed 858 patient records (2.1% with cancer) with imputation, encoding, and robust scaling.
- Trained a Gaussian Naive Bayes classifier; its 77.5% accuracy fell below the 97.3% baseline and it detected 1 of 7 cancer cases (ROC AUC 0.47 test, 0.61 cross-validated), showing that accuracy misleads on imbalanced clinical data.

WSU Phenomics Lab | USDA-NIFA REEU Program **Mt. Vernon, WA**
Data Science Intern *April 2024 - August 2024*
Advisors: Prof. Lisa DeVetter & Dr. Sindhuja Sankaran

- Analyzed color in 400 raspberries across 8 genotypes (Python, MATLAB); ANOVA found significant differences.
- Surveyed 15 people: 93% rated their color perception fair or good, yet scored only 13–53% on color tasks, showing that RGB-based phenomic tools can evaluate raspberry color for breeding more accurately than visual assessment.
- Fine-tuned an LLM chatbot (Phenomics Guide) to show stakeholders how phenomic technology can improve the precision and profitability of raspberry breeding; their confidence in the technology rose after use.

PRESENTATIONS

Willoughby, R., Sankaran, S., & DeVetter, L. Phenomics Tools in Raspberry Breeding: Color Perception and AI-Driven Strategies for Enhancing Breeder Adoption. *REEU Research Symposium*, Washington State University, Pullman, WA. August 1, 2024. Oral Presentation.

PROJECTS

SITE: Sustainable Infrastructure Trade-Off Evaluation
Research prototype, interactive tool

- Built a research prototype that estimates region-specific limits for AI data center deployment, weighing grid capacity, cooling, water stress, and carbon intensity.
- Combined hard-constraint checks with multi-objective trade-off analysis to evaluate proposed deployments.

U.S. Movie Revenue Case Study **Pullman, WA**
Course project, Probability & Statistics, WSU *December 2025*
R, regression

- Used confidence intervals and hypothesis testing to evaluate an industry profitability claim, rejecting the claim that average return on investment was 12%.
- Built multiple regression models on 2006 U.S. films to identify which characteristics (genre, rating, sequel status, release timing, distribution scale, critic scores) predicted opening-weekend and total box-office revenue.
- Refined the models to isolate key revenue drivers: opening theater count and sequel status predicted opening revenue, and critic scores explained about 80% of the variation in total domestic revenue.

ACCESSIBILITY AND LEADERSHIP EXPERIENCE

WSU Access Center **Pullman, WA**
Student Supervisor *August 2023 - December 2025*

- Remediated course materials and documents into accessible formats for students using assistive technology.
- Led recruitment and supervision of 20+ academic support specialists while coordinating accommodations for 150+ students per term, improving service quality, compliance, and timely access to resources.
- Led onboarding for new student staff, designing spreadsheet trackers and checklists to monitor hiring documentation and verify readiness for HR approval.
- Updated and maintained student records in the office database, which served ~2,000 students.

HONORS AND AWARDS

Merit Scholarship, PR[AI]RIE-PSAI (Paris School of AI), PSL Research University 2026
McCroskey Scholar | Washington State University 2025
President's Honor Roll | Washington State University 2024, 2025
Delegate, Harvard Project for Asian and International Relations, Harvard University, Cambridge, MA 2024

SKILLS

Programming & Data Processing: Python, MATLAB, SQL, R, Tableau, Excel
Computational Modeling and Analysis: ANOVA, Logistic Regression, Multi-objective optimization
Machine Learning: Supervised classification, model characterization and evaluation including LLMs and GNNs
Network & Behavioral Analysis: Social network analysis (Gephi), stakeholder analysis
Relevant Training: NVIDIA Fundamentals of Deep Learning, UC Davis Social Network Analysis, System Design, Presentation and Public Speaking